

Partitioned Probe Ptolemaic Graphs

David B. Chandler¹, Maw-Shang Chang², Ton Kloks, Sheng-Lung Peng³

¹Institute of Mathematics, Academia Sinica
Nangang, Taipei 11529, Taiwan
chandler@math.sinica.edu.tw

²Department of Computer Science and Information Engineering
National Chung Cheng University
Chiayi 62107, Taiwan
mschang@cs.ccu.edu.tw

³Department of Computer Science and Information Engineering
National Dong Hwa University
Hualien 974, Taiwan
lung@csie.ndhu.edu.tw

Abstract

Given a class of graphs $\mathcal{G}$, a graph G is a *probe graph of $\mathcal{G}$* if its vertices can be partitioned into two sets, $\mathbb{P}$ (the probes) and $\mathbb{N}$ (the nonprobes), where $\mathbb{N}$ is an independent set, such that G can be embedded into a graph of $\mathcal{G}$ by adding edges between certain nonprobes. If the partition of the vertices into probes and nonprobes is part of the input, then we call the graph a *partitioned probe graph of $\mathcal{G}$* . In this paper we show that there exists a polynomial-time recognition algorithm for partitioned probe graphs of the class of ptolemaic graphs.

1 Introduction

Given a partitioned graph $G = (\mathbb{P} + \mathbb{N}, E)$ with an *independent set* $\mathbb{N}$ of *nonprobes*, our aim is to find an embedding of G into a graph of a certain graph class $\mathcal{G}$ by adding some edges between nonprobes. This problem stems from a problem in resequencing pieces of DNA. There, some fault-tolerant recognition algorithm was desirable for the class of *interval graphs*. Since then research is under way on other graph classes as well.

A connected graph is *distance hereditary* if the distance between any two vertices remains the same in every connected induced subgraph [6]. We call a graph also distance hereditary if every component of the graph is such. A graph is *chordal* if every induced cycle with length greater than three

has a chord. A graph is *ptolemaic* if it is chordal and distance hereditary [7].

In this paper we show that there exists a polynomial-time algorithm that recognizes whether a partitioned graph is a partitioned probe ptolemaic graph or not.

2 Preliminaries

Let $G = (V, E)$ be a simple and undirected graph where V and E are the vertex and edge sets of G , respectively. Let $n = |V|$ and $m = |E|$. For a vertex x we write $N(x) = \{y \in V \mid (x, y) \in E\}$ for the set of *neighbors* of x in G and $N[x] = N(x) + x$ for its *closed neighborhood*. Let $d(x, y)$ denote the shortest distance from x to y .

Definiton 1. A pendant vertex in a graph is a vertex with exactly one neighbor.

Definiton 2. A twin in a graph is a module with two vertices. A twin is true if the vertices are adjacent. Otherwise the twin is false.

So, a false twin is a pair of nonadjacent vertices with exactly the same neighborhood and a true twin is a pair of adjacent vertices with the same closed neighborhood, where the closed neighborhood of a vertex is the set of neighbors plus the vertex itself.

It is a classical result that distance-hereditary graphs can be captured by forbidden induced subgraphs [1, 3]. For the house, hole, domino, and

gem, we refer to Figure 1. Note that, if a graph contains no induced house, hole, domino, nor gem, then adding a pendant vertex or creating a twin cannot give rise to one of these induced subgraphs.

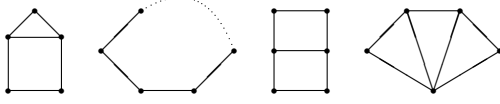


Figure 1: A house, a hole, a domino, and a gem.

Theorem 1 ([1]). *Let G be a graph. The following conditions are equivalent:*

1. G is connected and distance hereditary.
2. G is connected and does not contain a house, hole, domino, or gem as an induced subgraph.
3. Every connected induced subgraph of G with at least two vertices has a pendant vertex or a twin.
4. For every pair of vertices x and y with $d(x, y) = 2$, there is no induced x, y -path of length greater than two.

In this paper we consider *chordal* distance-hereditary graphs, i.e., ptolemaic graphs. A *chord* in a cycle is an edge of the graph which connects two vertices of the cycle that are not adjacent in the cycle. A cycle in a graph is *chordless* if it has no chord. A *hole* is a chordless cycle of length at least 5. Recall that a graph is *chordal* if it does not have any chordless cycle of length more than 3.

Originally, ptolemaic graphs were defined as follows.

Definiton 3 ([8]). *A connected graph is ptolemaic if for every four vertices x, y, u , and v ,*

$$d(x, y)d(u, v) \leq d(x, u)d(y, v) + d(x, v)d(y, u).$$

A graph is ptolemaic if every component is ptolemaic.

Ptolemaic graphs can be characterized as those chordal graphs in which every 5-cycle has at least three chords, or, as those chordal graphs in which all chordless paths are shortest paths [2, 5, 9]. We will use the following characterization of ptolemaic graphs.

Theorem 2 ([7]). *A graph is ptolemaic if and only if it is distance hereditary and chordal.*

Corollary 1. *Notice that holes, houses, and dominos are not chordal. Hence ptolemaic graphs are those chordal graphs which are gem-free.*

3 Recognition of partitioned probe ptolemaic graphs

We show that there exists a polynomial-time algorithm which checks whether a graph G with a given partition of the vertices into a set $\mathbb{P}$ of probes and an independent set $\mathbb{N}$ of nonprobes can be embedded into a ptolemaic graph by adding edges between some nonprobes. We show that the partitioned probe ptolemaic graphs have a nice characterization resembling the third item of Theorem 1 that characterizes distance-hereditary graphs. We need the following lemma. Recall that a *simplicial vertex* is a vertex whose neighborhood is a clique.

Lemma 1. *Assume G is ptolemaic and let x be a vertex in G . The graph F obtained from G by adding edges such that $N(x)$ becomes a clique is also ptolemaic.*

Proof. It is easy to see that making a clique of $N(x)$ does not create a chordless cycle of length more than 3 since G is chordal. Assume that F has a gem S . If x is a vertex of S then it must be simplicial. Deleting the edge between the two neighbors in S creates a chordless cycle. If x is not in S then it is adjacent to a triangle or an edge in S . Then deleting the edge or one or more edges of the triangle creates a chordless 4-cycle or leaves a gem with vertex x . $\square$

We call a subgraph a *probe clique* if a pair of vertices is adjacent whenever at least one of them is a probe.

Definiton 4. *Let $G = (\mathbb{P} + \mathbb{N}, E)$ be a partitioned graph. A pair of vertices $\{x, y\}$ is a ptolemaic twin if one of the following holds.*

1. $x, y \in \mathbb{P}$, and
 - i. $\{x, y\}$ is a module in G , and
 - ii. if x and y are not adjacent then $N(x)$ is a probe clique, or
2. $x, y \in \mathbb{N}$ and $\{x, y\}$ is a module, or
3. $x \in \mathbb{P}$ and $y \in \mathbb{N}$, and
 - i. $N(y) - x = (N(x) - y) \cap \mathbb{P}$, and
 - ii. if x and y are not adjacent then $N(x)$ is a probe clique.

Our main result is the following characterization.

Theorem 3. *A partitioned graph $G = (\mathbb{P} + \mathbb{N}, E)$ is partitioned probe ptolemaic if and only if every connected induced subgraph of G with at least two vertices has a pendant vertex or a ptolemaic twin.*

Proof. Assume G is partitioned probe ptolemaic, and let H be an embedding of G . Let C be the vertex set of a connected induced subgraph of G and assume $|C| \geq 2$. By Theorem 2, the class of ptolemaic graphs is hereditary; thus $H[C]$ is also ptolemaic. If x is a pendant vertex in $H[C]$, then since $G[C]$ is connected, x is also a pendant vertex of $G[C]$. Suppose $H[C]$ has a twin, say $\{x, y\}$. Notice that, if x and y are not adjacent, then $N_H(x) = N_H(y)$ is a clique, since the graph is chordal. It is easy to check that $\{x, y\}$ is a ptolemaic twin in $G[C]$.

Assume that every connected induced subgraph of G has a pendant vertex or a ptolemaic twin. We may assume that G is connected. Let x be a pendant vertex in G . By induction $G - x$ has an embedding as a ptolemaic graph. Since a gem or chordless cycle has no pendant vertex we may add x to the embedding.

Suppose G has a ptolemaic twin $\{x, y\}$ with $x, y \in \mathbb{P}$. By induction, the graph $G - y$ can be embedded as a ptolemaic graph. If x and y are adjacent then we add y as a twin of x to this embedding. This creates no chordless cycle nor any gem. If x and y are nonadjacent, then by Lemma 1 we may make x simplicial in this embedding. We add y as a twin of x .

Suppose G has a ptolemaic twin $\{x, y\}$ with $x, y \in \mathbb{N}$. By induction, the graph $G - y$ has an embedding into a ptolemaic graph. We add y to the embedding as a true twin of x . This cannot create a gem or a chordless cycle.

Suppose G has an ptolemaic twin with $x \in \mathbb{P}$ and $y \in \mathbb{N}$. Consider an embedding of $G - y$. If x and y are connected we can add y as a true twin of x . Assume x and y are not connected. Let F be an embedding of $G - y$. Then by Lemma 1 we may make a clique of $N_F(x)$. Add y as a false twin of x . $\square$

We now easily obtain our recognition algorithm as follows. For each iteration, we first find a pendant vertex x or a ptolemaic twin $\{x, y\}$. In case that at least one vertex in the ptolemaic twin $\{x, y\}$ is nonprobe, we assume that x is the nonprobe. Then we remove x from G and go to the next iteration. If there is an iteration such that we cannot find a pendant vertex or a ptolemaic twin, then, by Theorem 3, G cannot be probe

ptolemaic. It is not hard to see that a pendant vertex can be found in linear time. For the implementation of finding a ptolemaic twin, we use the adjacency matrix. The algorithm maintains four neighborhood adjacency matrices $\mathcal{S}^i$, $1 \leq i \leq 4$, of vertices where $\mathcal{S}^1 = \{N[x] \mid x \in \mathbb{P}\}$, $\mathcal{S}^2 = \{N(x) \mid x \in \mathbb{P}\}$, $\mathcal{S}^3 = \{N[x] \cap \mathbb{P} \mid x \in \mathbb{P} + \mathbb{N}\}$, and $\mathcal{S}^4 = \{N(x) \cap \mathbb{P} \mid x \in \mathbb{P} + \mathbb{N}\}$. It is easy to see that a ptolemaic twin can be determined if one of the four matrices has two vertices x and y that have the same neighborhood and if x is a probe vertex and x is not adjacent to y , then $N(x)$ must form a probe clique. For each matrix, we use a radix sort to sort the vertices according to their adjacency relation to find a possible ptolemaic twin. This work can be done in $O(n^2)$ time. Further, if it is the case that we need to check a probe clique, we then check it. This checking process can be done in $O(n + m)$ time. Finally, we update the matrices after we remove a vertex. A simple way to do this is to change all the entries corresponding to the removed vertex from 1 to 0. Making these updates takes $O(n)$ time. Thus, the total time is bounded in $O(n^3)$. Therefore, we have the following theorem.

Theorem 4. *There exists an $O(n^3)$ algorithm for the recognition of partitioned probe ptolemaic graphs.*

4 Concluding remarks

In this paper, we propose the first polynomial-time algorithm to recognize partitioned probe ptolemaic graphs. One of the open problems is the following. Recall that a *strongly chordal graph* is a chordal graph without a sun [4]. A *sun* is the graph obtained from an even cycle with at least six vertices, by making a clique of a maximum independent set in the cycle. If the cycle has length $2k$, the sun is called a k -sun. As an example see Figure 2. This class is an important subclass of the chordal graphs for which, *e.g.*, the DOMINATING SET problem is polynomial. This problem remains NP-complete when restricted to the class of chordal graphs. Notice that, since every sun contains a gem, the class of ptolemaic graphs is properly contained in the class of strongly chordal graphs.

Conjecture 1. *The recognition problem of partitioned probe strongly chordal graphs is polynomial.*

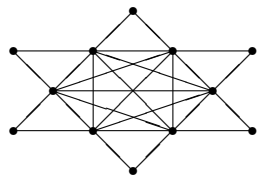


Figure 2: A 6-sun.

References

- [1] Bandelt, H. J. and H. M. Mulder, Distance-hereditary graphs, *Journal of Combinatorial Theory, Series B* **41** (1986), pp. 182–208.
- [2] Brändstadt, A., V. B. Le, and J. P. Spinrad, *Graph classes: A survey*, SIAM Monographs on Discrete Mathematics and Applications, Philadelphia, 1999.
- [3] D’Atri, A. and M. Moscarini, Distance-hereditary graphs, Steiner trees, and connected domination, *SIAM J. Comput.* **17** (1988), pp. 521–538.
- [4] Farber, M., Characterizations of strongly chordal graphs, *Discrete Mathematics* **43** (1983), 173–189.
- [5] Farber, M. and R. E. Jamison, Convexity in graphs and hypergraphs, *SIAM J. Alg. Discrete Methods* **7** (1986), pp. 433–444.
- [6] Howorka, E., A characterization of distance-hereditary graphs, *The Quarterly Journal of Mathematics* **28** (1977), pp. 417–420.
- [7] Howorka, E., A characterization of ptolemaic graphs, *Journal of Graph Theory* **5** (1981), pp. 323–331.
- [8] Kay, D. C. and G. Chartrand, A characterization of certain ptolemaic graphs, *Canad. J. Math.* **17** (1965), pp. 342–346.
- [9] Soltan, V. P., d -convexity in graphs, *Soviet Math. Dokl.* **28** (1983), pp. 419–421.